

Applications to probability



1 $\frac{1}{2}$

2 $\frac{253}{9996}$

3 $\frac{1}{10}$

4 $\frac{12}{35}$

5 $\frac{42}{143}$

6 a 15

b $\frac{1}{15}$

7 a 10

b $\frac{2}{5}$

8 a 24

b $\frac{1}{24}$

c 8 meals

9 a 792

b $\frac{1}{12}$

c $\frac{1}{220}$

10 a 20

b $\frac{1}{20}$

11 a 126

b 35

c $\frac{5}{18}$

12 a 20

b $\frac{1}{2}$

13 a 1287

b 14



14

a 16

b $\frac{1}{16}$

c $\frac{1}{16}$

d $\frac{1}{4}$

15

a 325

b $\frac{1}{325}$

16

a 1.01%

b 35.35%

c 2.02%

17

a 5.52%

b 0.01%

c 0.07%

18

a $\frac{429}{5168}$

b $\frac{4739}{5168}$

19

a $\frac{4}{15}$

b $\frac{22}{65}$

20

a $\frac{91}{204}$

b $\frac{113}{204}$

21

a $\frac{4}{143}$

b $\frac{49}{143}$

22

a $\frac{1}{1287}$

b $\frac{25}{1287}$

c $\frac{925}{1287}$

d $\frac{56}{1287}$

23

a $\frac{66}{195\,755}$

b $\frac{83\,486}{195\,755}$

24

a 28 combinations

b $\frac{1}{4}$

c $\frac{1}{4}$

25

a $\frac{1}{20}$

b $\frac{2}{5}$

c $\frac{1}{5}$

d $\frac{1}{5}$

26 a $\frac{1}{9}$

b $\frac{5}{9}$



c $\frac{1}{3}$

27 a $\frac{3}{5}$

b $\frac{4}{5}$

c $\frac{3}{5}$

28 $\frac{2}{1885}$

29 a $\frac{48}{95}$

b $\frac{2}{171}$

c $\frac{8}{15}$

d $\frac{48}{143}$

30 a $\frac{61}{91}$

b $\frac{95}{406}$

31 a $\frac{5}{9}$

b $\frac{4}{9}$

c $\frac{5}{6}$

32 a $\frac{15}{28}$

b $\frac{75}{281}$

33 a $\frac{99}{754}$

b $\frac{11}{68}$

c $\frac{9}{133}$

34 a $\frac{4}{9}$

b $\frac{5}{9}$

c $\frac{4}{13}$

35 a $\frac{2035}{3876}$

b $\frac{1051}{104\,006}$

36 a 110

b $\frac{2}{11}$

c $\frac{2}{7}$

37 a $\frac{1}{45}$

b $\frac{8}{45}$

38

a $\frac{2}{11}$
e $\frac{1}{11}$

b $\frac{9}{11}$
f $\frac{1}{11}$



c $\frac{7}{11}$

d $\frac{5}{11}$

39

a 1

b $\frac{2}{3}$

c $\frac{1}{2}$

d $\frac{2}{k-1}$